MARKET INTELLIGENCE · CASE STUDY

Which skills are rising, which are fading, and what does it pay to know?

An analysis pipeline on a **simulated** dataset: 50,000 job postings · 500 companies · 10 countries · 25 skills · 2021–2025.

Python | SQL | LLM APIs | Power BI | dbt | Cloud | VBA/Macros

SIMULATED DATA · PIPELINE & METHOD DEMO, NOT MARKET RESEARCH

50,000	10	25	2021–25	+5,323%
JOB POSTINGS	COUNTRIES	SKILLS	TIME RANGE	LLM APIs GROWTH (MODEL)

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THE STORY IN NUMBERS

What changed between 2021 and 2025

+5,323%

LLM APIs demand: from 1.1% to 58.6% of postings.

~2.7 yrs

VBA half-life. Demand fell from 31.0% to 7.8% (linear model).

91.3%

Python demand in 2025, up from 46.0%

89.5%

SQL in 2025, essentially flat (88.0% in 2021)

+22%

Salary premium assumed for LLM APIs

In this data, LLM APIs went from almost nothing in 2021 to more than half of postings in 2025.

FIVE THINGS THE ANALYSIS SHOWS

01 · LLM APIs: the fastest-growing skill by a distance

02 · SQL stays near 89.5% the whole time

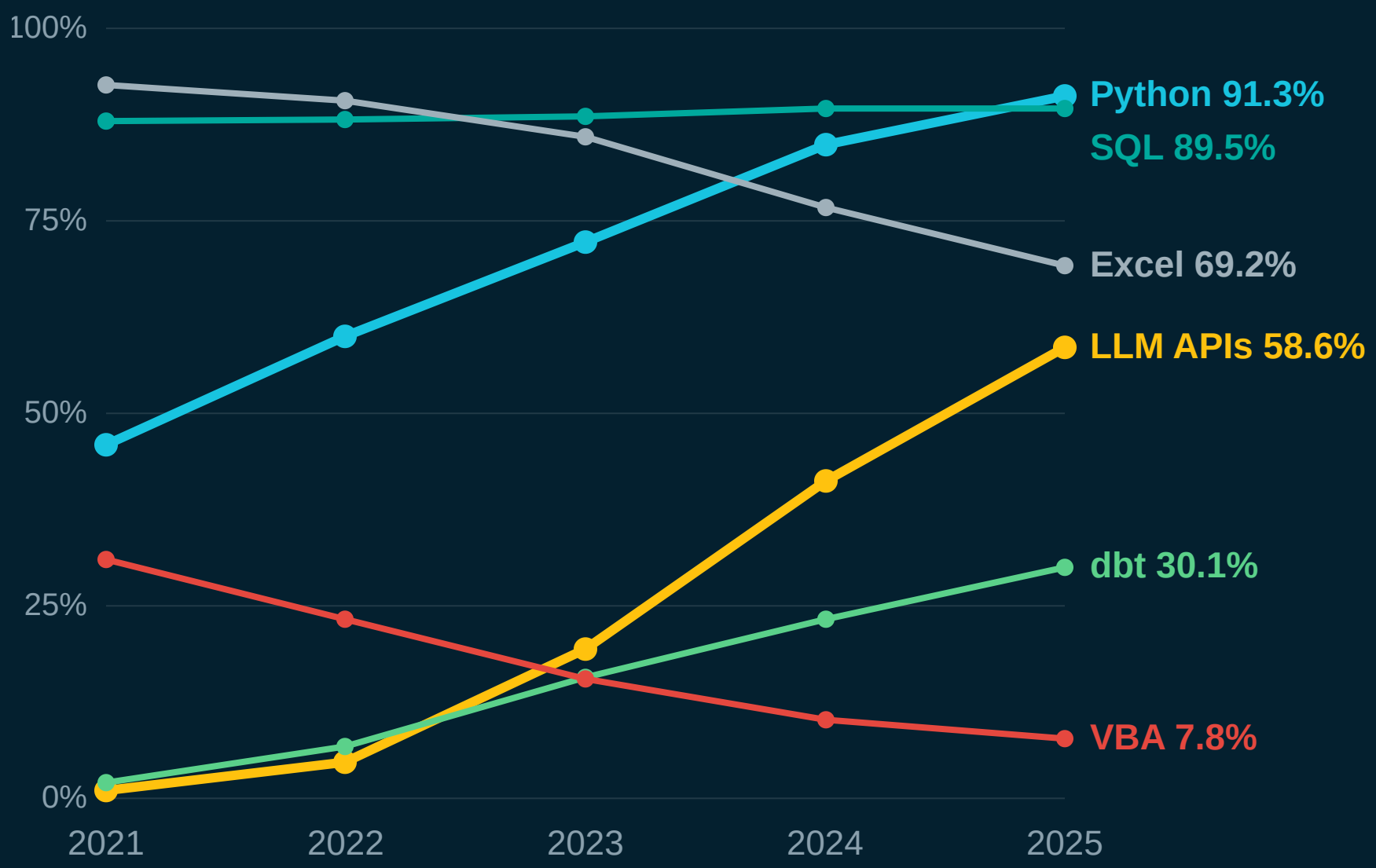
03 · VBA: shortest half-life (2.7 yrs), so migrate to Python

04 · dbt: 2.1% to 30.1%, the modern data stack

05 · AI/ML skills carry a 13–22% premium (a model input)

SKILL DEMAND TRENDS 2021–2025

How skill demand moved 2021 to 2025



Python rose steadily to 91.3%. LLM APIs went from 1.1% to 58.6%, the biggest move in the data.

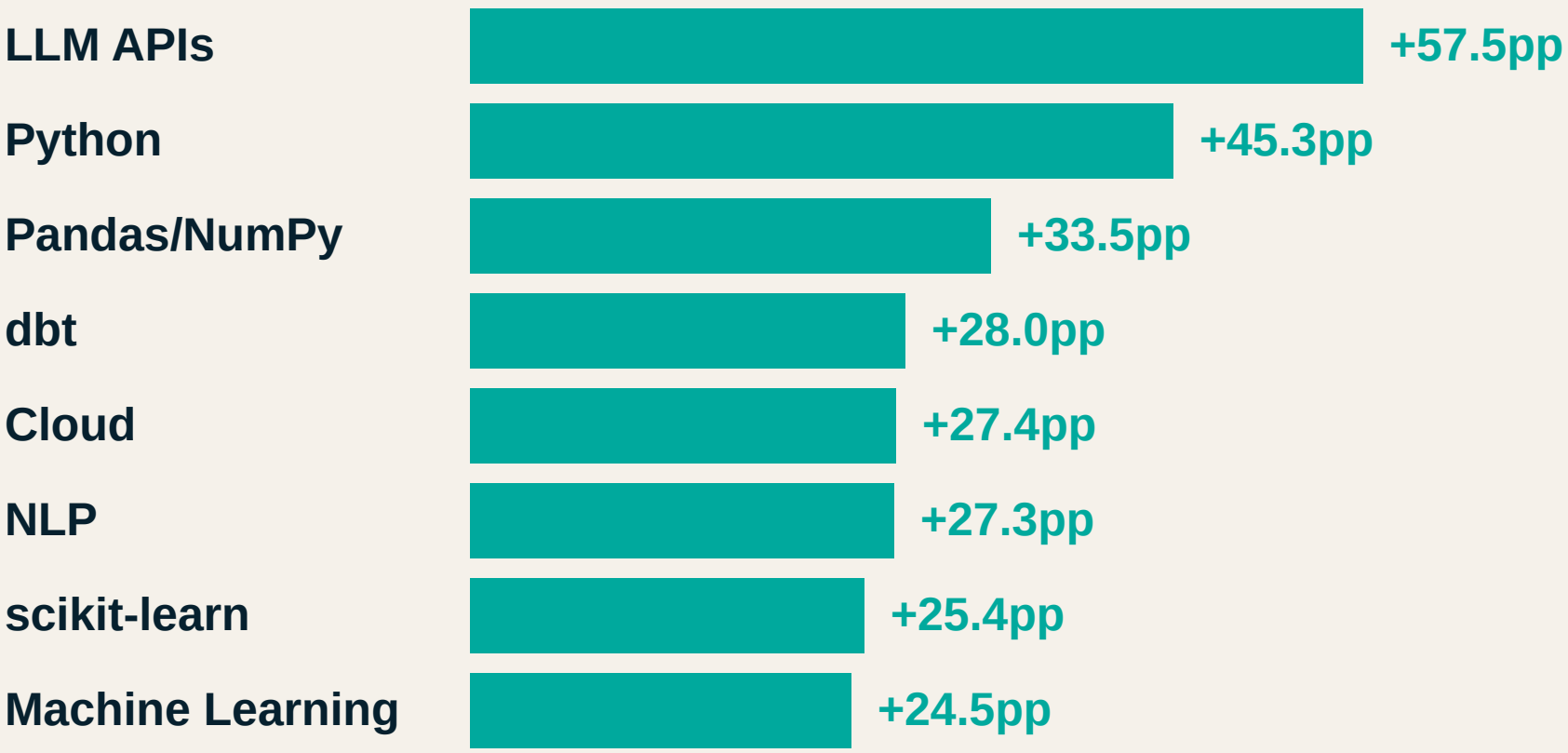
2025 DEMAND SNAPSHOT

Python	SQL	LLM APIs	Cloud	dbt	VBA
91.3%	89.5%	58.6%	53.1%	30.1%	7.8%

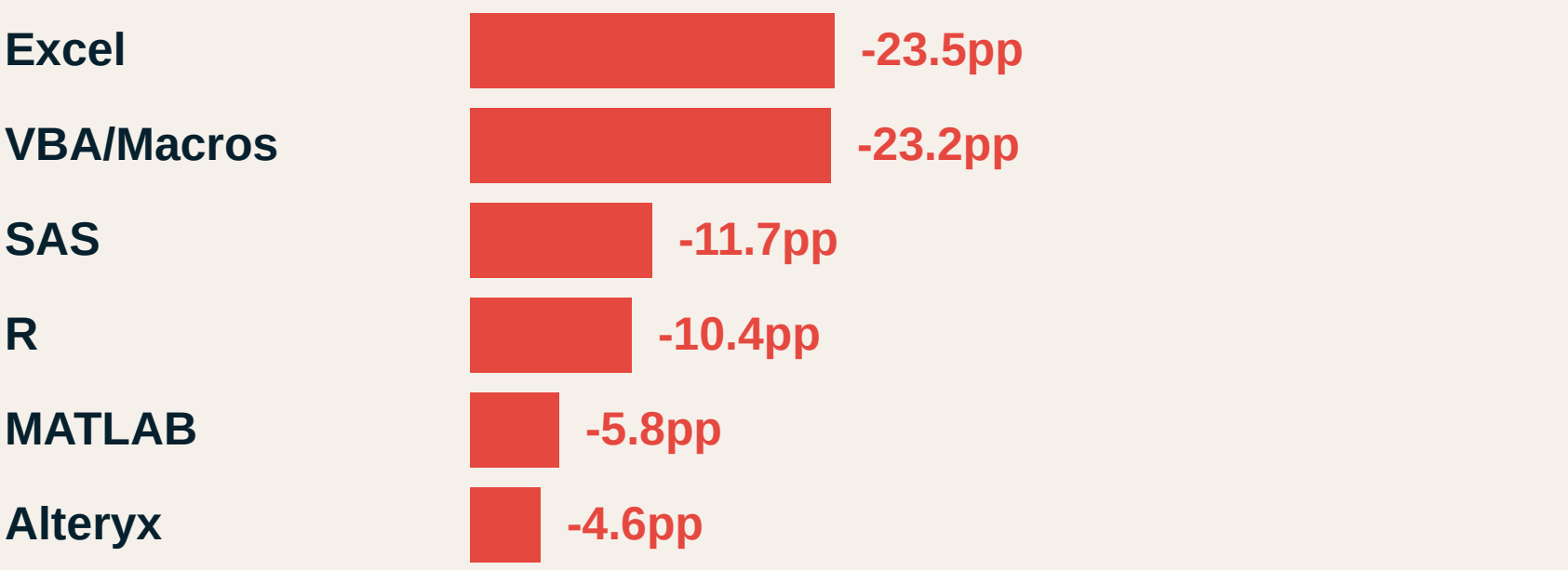
SKILL GROWTH RANKINGS

Biggest rises and falls

FASTEST GROWING (pp change 2021–2025)



FASTEST DECLINING (pp change 2021–2025)



LLM APIs and Python grow the most. Excel and VBA fall the most.

COUNTRY COMPARISON · 2025

USA vs India on LLM APIs, 2025

	Python	SQL	LLM APIs	ML	Cloud	dbt
USA	91.2%	90.8%	66.2%	65.1%	53.2%	31.9%
Singapore	91.4%	86.9%	64.1%	62.2%	50.6%	30.5%
UK	90.2%	90.7%	62.5%	61.5%	51.1%	32.1%
Australia	92.9%	87.6%	57.9%	61.1%	55.4%	29.0%
Germany	92.1%	90.7%	56.6%	57.6%	53.3%	30.4%
India	90.5%	90.2%	53.8%	54.6%	52.7%	26.1%

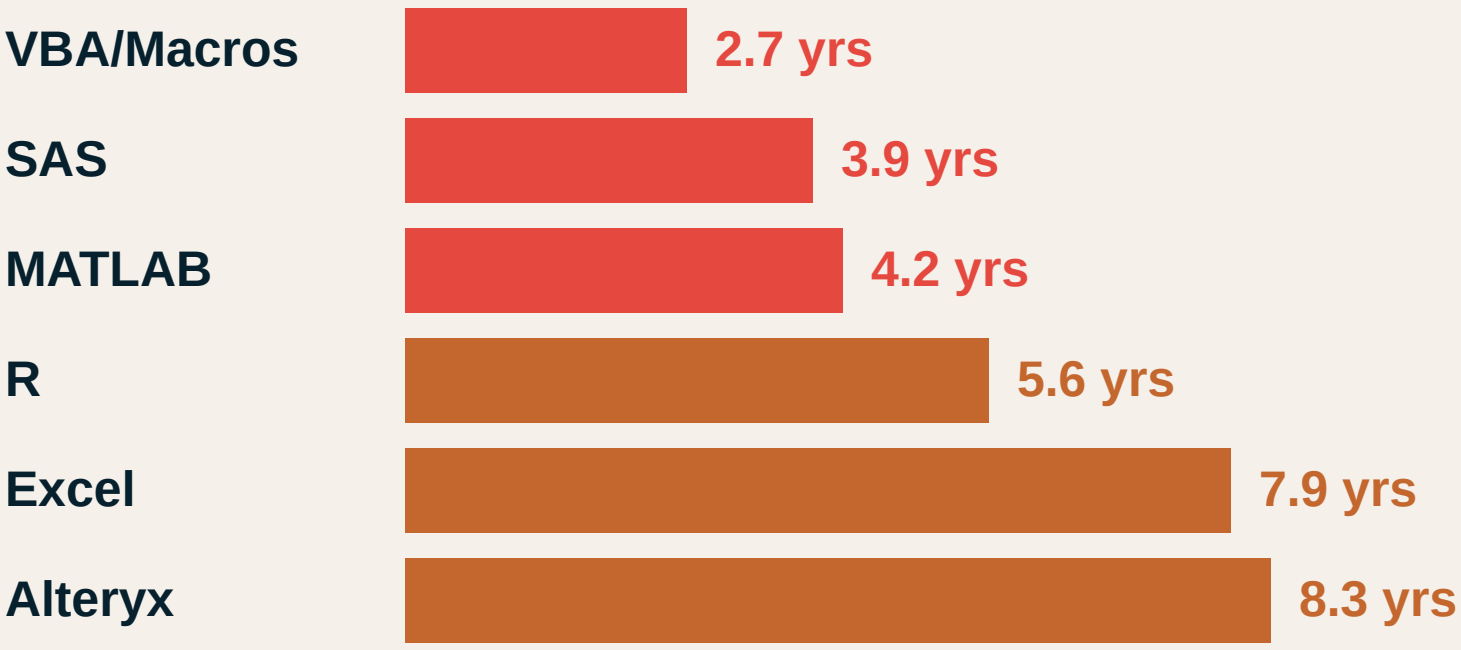
> USA leads LLM APIs demand at 66.2% of postings. India sits at 53.8%. The 12.4pp gap comes from country multipliers set in the generator, so read it as an illustration of the method, not an observed adoption lag.

Cloud gap USA vs India: 0.5pp (Cloud demand has no country multiplier in the model).

SKILL HALF-LIFE INDEX

How long until demand halves

YEARS UNTIL DEMAND HALVES (LINEAR MODEL)



Half-life = years for demand to fall to half of its 2021 level at the 2021–2025 rate. VBA (31.0% to 7.8%) is the shortest at 2.7 years. Excel (7.9 years) is declining, not dead.

HOW IT IS CALCULATED

annual_change = (demand_2025 − demand_2021) / 4

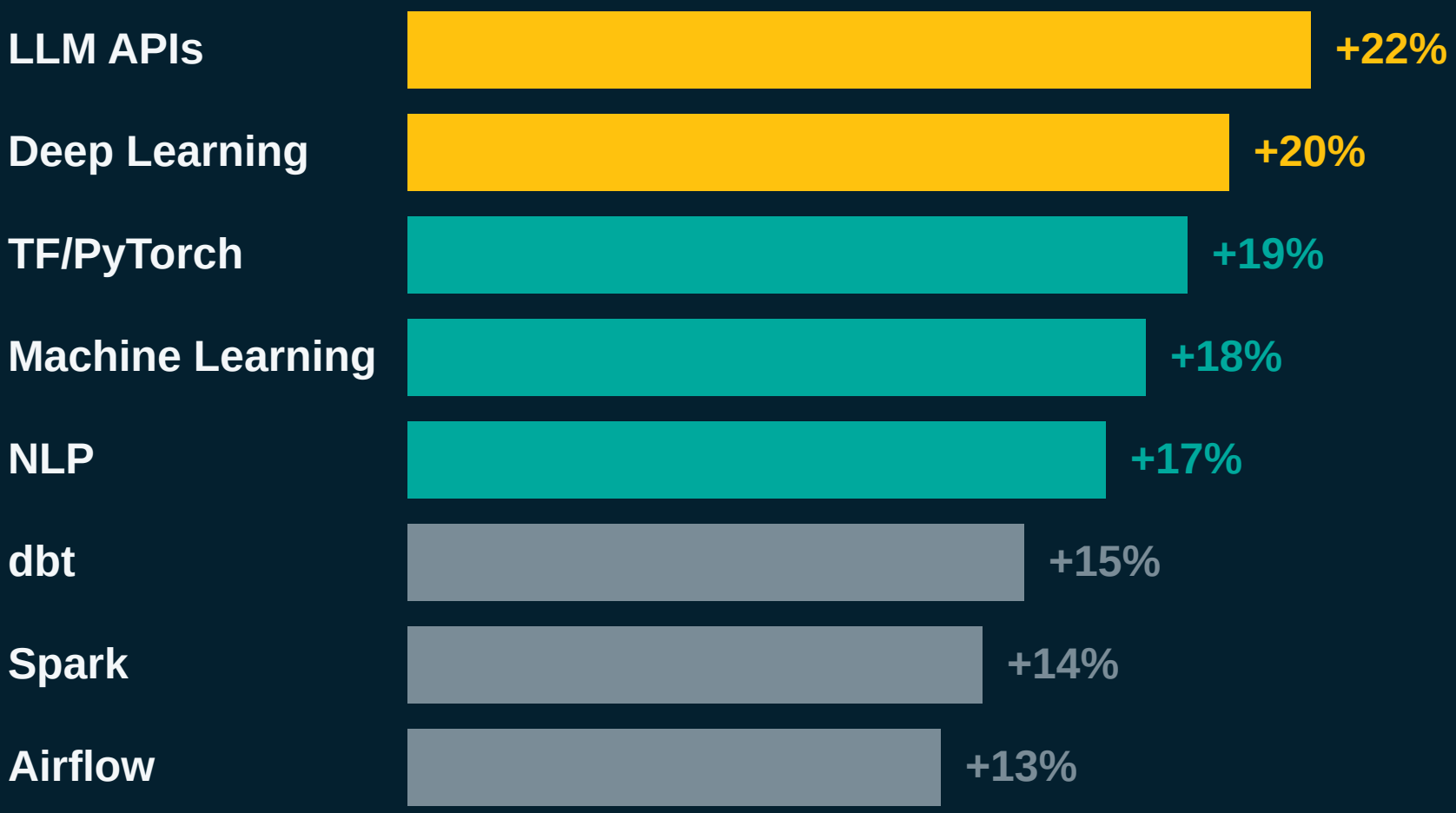
half_life = demand_2021 / (2 × |annual_change|)

A planning heuristic from two endpoints, not a forecast.

VBA: stop investing. **SAS, MATLAB:** plan a transition. **Excel:** maintain, don't deepen.

Salary premium by skill (assumed)

PREMIUM VS BASELINE ANALYST SALARY

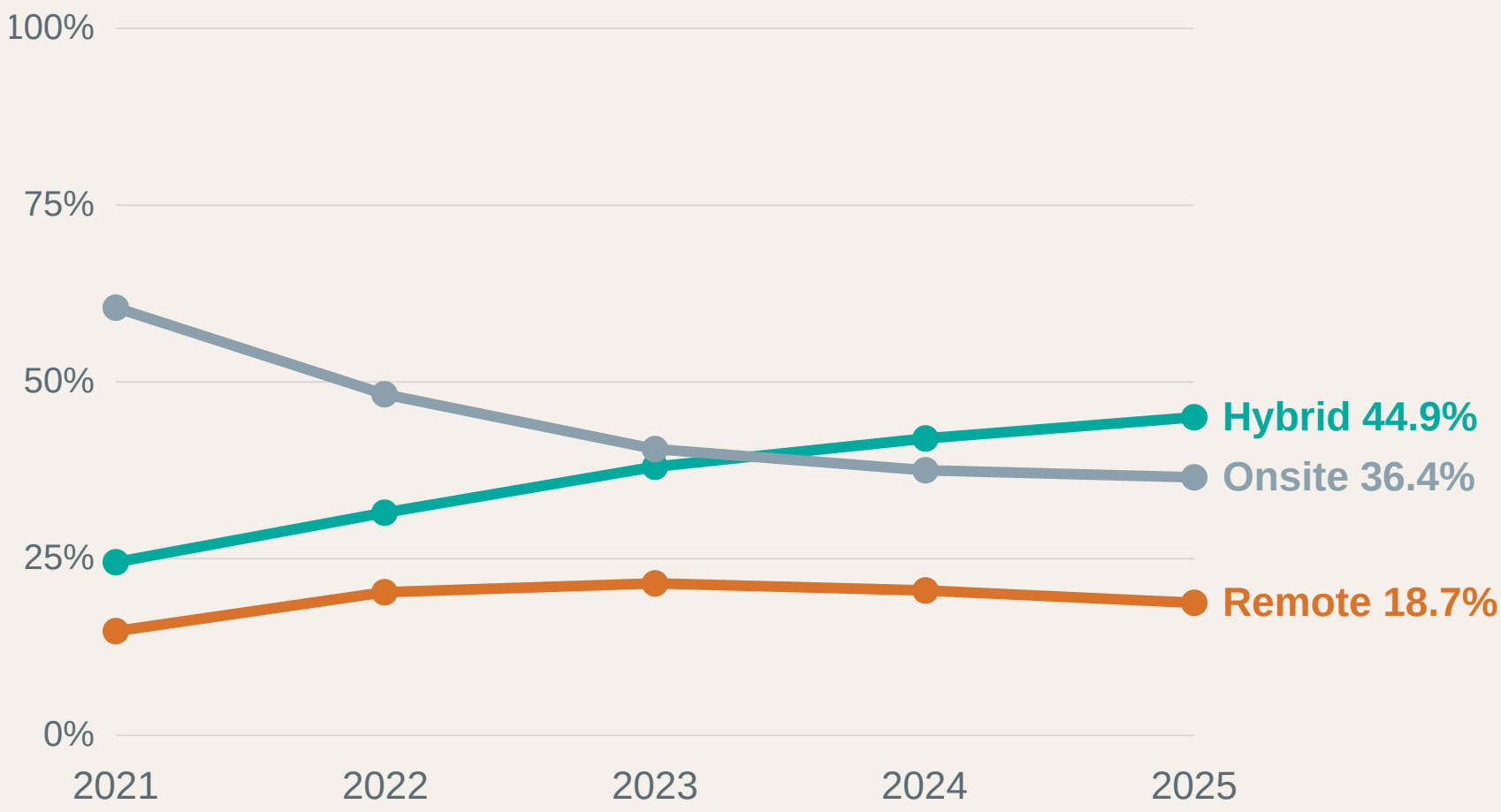


> On a 10 lakh base, a 22% premium for LLM APIs is 2.2 lakh; Deep Learning (20%) is 2.0 lakh; Machine Learning (18%) is 1.8 lakh.

Important: these premiums are fixed assumptions in the data generator, not estimates from real salary data. A production version would estimate them by regressing salary on skills, controlling for country, experience and sector.

REMOTE WORK TREND

Remote, hybrid and onsite



Hybrid rose from 24.6% to 44.9%. Onsite fell from 60.6% to 36.4%. Remote peaked at 21.6% in 2023 and eased to 18.7%.

2025 BREAKDOWN

18.7%	44.9%	36.4%
Remote	Hybrid	Onsite

FULL PROJECT ON GITHUB

The project on GitHub

10 MySQL queries · 6 Python charts · skill half-life index · interactive dashboard · Power BI build guide.

Python · Pandas | SQL · MySQL | Matplotlib · Seaborn | Power BI · DAX

25 skills | 10 countries

PROJECT SUMMARY

Data & MySQL
job_generator.py + load_mysql.py build 5 MySQL tables with 440k+ records

Analysis
analysis.py runs SQL on MySQL and writes 6 charts incl. the half-life index

SQL
mysql_queries.sql: trends, half-life, co-occurrence, salary benchmark

Dashboard
4-page interactive HTML dashboard + Power BI build guide with DAX

github.com/Ajaypatel06

The dataset is simulated, with assumptions documented in the README. Open to Data Analyst / Product Analyst roles · Mumbai. Comment or message me if you want to talk about it.